

1 Bayesian Normal Linear Regression

1.1 Foundations and Likelihood

The goal is to assess associations between a response Y and regressors X or predict new observations Y' . In the Bayesian framework, we assume response normality where $Y_i \sim N(\mu_i, \sigma^2)$. μ_i represents the **systematic component** of the regression model, and σ^2 is the common variance shared by all responses.

```
# OLS vs Bayesian Comparison
# OLS can produce negative intercepts that lack physical sense.
# Bayesian modeling can fix this via restricted prior support.
```

1.2 Stan Program Blocks and Prior Specifications

Stan utilizes specific blocks: **data**, **parameters**, and **model**. The **generated quantities** block is executed **after** obtaining posterior samples and is used for predictive inquiries.

Posterior variability in parameters: is the variability from approximate joint posterior distribution. These predictions will be computed using the fixed values **pred.x.1** and **pred.x.2**.

Sampling variability: Normal random number generator `normal_rng()` is added for random noise in MCMC

- **vector:** Linear-algebra type with operator support; preferred for dot products and matrix multiplication like $X * \beta$.
- **array:** Container type best for indexing, nested structures, and element-by-element loops.
- **int:** Integer data **must** be declared as arrays.

```
model {
  #Priors allow for clever modeling by choosing appropriate support for parameters.
  beta_0 ~ gamma(7.5, 1); #for Intercept A Gamma prior is used if the response is naturally non-negative.
  beta_1 ~ normal(0, 1000); #for coeff Typically assigned a Normal prior and A large variance reflects prior uncertainty regarding associations.
  sigma ~ inv_gamma(0.001, 0.001); #The Inverse Gamma distribution is a popular non-negative choice.
  y ~ normal(beta_0 + beta_1 * x_1, sigma);
}
```

```
library(bayesrules)
summarize_beta_binomial(a = 4, b = 12, y = 20, n = 200)
```

2 Bayesian Hypothesis Testing

2.1 One-Sided Tests and Odds

Unlike frequentist testing, each hypothesis (H_0 and H_a) is assigned a **probability** derived from the posterior distribution.

- **Posterior Odds:** The ratio $P(H_a|Y)/P(H_0|Y)$.
- **Prior Odds:** The ratio $P(H_a)/P(H_0)$.

2.2 The Bayes Factor

The **Bayes Factor** is the substitute for the p-value. It compares posterior odds to prior odds to see how inference changed given evidence. $BF = 1$: Plausibility of H_a remained the same. $BF > 1$: Plausibility of H_a increased. $BF < 1$: Plausibility of H_a decreased.

```
posterior_prob_H_0 <- pbeta(0.15, 24, 192)
posterior_prob_H_a <- pbeta(0.15, 24, 192, lower.tail = FALSE)
posterior_odds <- posterior_prob_H_a / posterior_prob_H_0
prior_prob_H_0 <- pbeta(0.15, 4, 12)
prior_prob_H_a <- pbeta(0.15, 4, 12, lower.tail = FALSE)
prior_odds <- prior_prob_H_a / prior_prob_H_0
#1/prior_odds times as plausible as the H_a that this mean exceeds
bayes_factor <- posterior_odds / prior_odds
round(qbeta(c(0.025, 0.975), 24, 192), 2) #2 sided test If the 95%
credible interval falls within this range, there is
reasonable evidence for H_0.
```

2.3 Ethics: B-Hacking

B-hacking involves changing priors after seeing data to make results look decisive. To avoid this, commit to priors beforehand and perform **prior sensitivity checks** by refitting the model with slightly stronger/weaker priors to ensure stable conclusions.

3 Bayesian Binary Logistic Regression

Bayesian regression can be extended to **GLMs**, specifically for binary responses. The key modeling term is the **probability of success** (π_i), assumed to follow a *Bernoulli*(π_i) distribution.

3.1 The Link Function

To relate the binary response to regressors, we use the **link function**, which is the **log of the odds**. It is defined as: $log(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \beta_1 X_{i,1} + \dots + \beta_k X_{i,k}$. The systematic component allows parameters to be unrestricted on the real scale while the probability π_i remains in $[0 - 1]$ via the **sigmoid function**.

3.2 Prior Predictive Distribution

A **prior-only model** uses Monte Carlo simulation (not MCMC) to draw from prior distributions without data. This allows us to visualize **prior sigmoid functions**, which represent what we believe is plausible before seeing evidence. Prior predictive distributions for probabilities often appear non-sensical or extreme before being updated by the likelihood.

3.3 Stan Program Blocks and Prior Specifications

data: Includes the training size n , the vector of predictors, and the array of binary responses. **parameters:** Declares the regression coefficients β_0 and β_1 . **model:** Specifies the priors and uses `bernoulli_logit()` to indicate the logit link function for the likelihood. The **generated quantities** block creates derived values after sampling. `inv_logit()` is used to transform log-odds back into the probability scale ($\hat{\pi}$). To simulate future binary outcomes (0/1), a **Bernoulli random number generator** (`bernoulli_rng`) is applied to the predicted probability.

```
model {
  and beta_1 (slope) typically receive
  beta_0 ~ normal(0, 100); #Parameters beta_0 (intercept) and
    beta_1 (slope) Normal priors
  beta_1 ~ normal(0, 100); #mean of zero ($\mu=0$) is often assumed
    to reflect prior neutrality regarding associations
  y ~ bernoulli_logit(beta_0 + beta_1 * x); #large variance makes
    the prior weakly informative
} #bernoulli_logit handles log-odds
generated quantities {
  real pi_pred;
  int y_pred; pi_pred = inv_logit(beta_0 + beta_1 * income_pred);
  y_pred = bernoulli_rng(pi_pred);
}
```

3.4 Inference and Comparison

Credible Intervals vs. P-values— In a Bayesian framework, we do not use **p-values**. Instead, we use **posterior credible intervals** to test if a coefficient is different from zero. If the 95% credible interval does not contain zero, we conclude the regressor is associated with the log-odds of the response.

3.5 Bayesian vs. Frequentist (OLS, GLM)

OLS—A Non-negative Bayesian intercept is more precise than the negative OLS estimation. Point estimates differ greatly, but the Bayesian one is more precise. Both approaches show considerable variability.

GLM—When using weakly informative priors, Bayesian posterior means are often **practically equal** to frequentist Maximum Likelihood Estimates (MLE) from `glm()`. For example, an estimate for β_1 of 0.009 implies that for each unit increase in the regressor, the subject is $exp(0.009) \approx 1.01$ times more likely to be a "success".

3.6 Posterior Predictive Inference

Eg., our posterior mean indicates that for each one thousand USD increase in income, a subject is 1.01 times more likely to believe in climate change than not to.

4 Bayesian Hierarchical Modeling

Standard Bayesian models involve a likelihood conditioned on parameters with fixed hyperparameters. **Hierarchical models** make these hyperparameters random variables with their own priors. This allows the model to account for two sources of variability: **Within-group variability** (differences among observations in a group) and **Between-group variability** (differences from one group to another). It is the best of complete pooled and non-pooled models.

4.1 Modeling Approaches Comparison

- **Complete Pooled:** Combines all data into one pool to estimate an overall probability. This ignores individual group differences, potentially leading to high bias.
- **Non-Pooled:** Estimates parameters for each group separately. While flexible, it produces noisy or "wild" estimates for groups with small sample sizes.
- **Hierarchical:** Assumes each group has its own rate, but these rates are drawn from a shared **population distribution**.

4.2 Borrowing Strength

Key property of hierarchical models. Groups with little data "lean" toward the overall pattern of the population, while data-rich groups remain mostly unchanged. This shared information stabilizes estimates and reduces posterior variance compared to non-pooled models.

Prior Specifications In a hierarchical rocket success model, success probabilities π_i follow a *Beta*(a, b) distribution. The parameters a and b are assigned **Gamma priors** (Chi-squared also) (e.g.,

Gamma(0.001, 0.001)), which are continuous and non-negative, reflecting the support required for Beta shape parameters.

The equation $\alpha_h + \beta_1 x_{h,i} + \beta_2 x_{h,i}^2 + \beta_3 x_{h,i}^3$ can be dissected as each region has its own intercept, but all regions share the same slopes. region-specific intercepts are not estimated completely separately from one another. Instead, they are assumed to come from a common population-level distribution, so information is shared across regions. This creates what it is called partial pooling, where regions with limited data are pulled somewhat toward the overall population pattern rather than being estimated in isolation. The fitted regional relationship can be overly sensitive to limited information rather than reflecting a broader underlying pattern. A hierarchical regression model helps by borrowing strength from the other regions through the shared population distribution. As a result, the region's estimates are typically more precise, less sensitive to random variation from a small sample, and often less extreme than they would be under no pooling.

4.3 Inference and Prediction

Posterior Results: Hierarchical models often yield more **precise results** with narrower credible intervals than non-pooled models. The model learns the average population probability by analyzing the posterior means of the hyperparameters (a and b) through a theoretical Beta distribution.

Predicting for New Groups: To predict the probability π_{pred} for a entirely **new group** not in the original dataset, the **generated quantities** block uses the `beta_rng(a, b)` function. This prediction reflects the population mean and the best Bayesian estimate possible without specific group covariates.

4.4 Relationship to Frequentist Methods

Hierarchical Bayesian models are the counterpart to frequentist **mixed-effects models**. They allow for generalization to new observations and the ability to use information from the entire dataset to infer group-specific probabilities.

5 MCMC Diagnostics: Visual Tools

5.1 Trace Plots and Density Overlays

Trace plots illustrate posterior sampling stability; they should show a flat horizontal pattern with no trends or "stuck" chains. **Empirical density plots** check consistency across multiple chains; overlaid plots should show similar centers, spreads, and shapes for the same parameter. Large discrepancies in density suggest chains are sampling from different posterior regions.

6 MCMC Diagnostics: Quantitative Metrics

6.1 Effective Sample Size (N_{eff})

N_{eff} measures the number of **independent samples** required to achieve an accurate approximation. Because MCMC draws are autocorrelated, N_{eff} is often smaller than the raw number of draws N . The ratio N_{eff}/N should ideally be close to 1; a ratio < 1 suggests a need for more iterations or thinning.

6.2 Autocorrelation

Autocorrelation measures how strongly a draw relates to previous ones. A desirable pattern is for autocorrelation to **drop quickly** as lag increases (e.g., before lag 5). Slowly decaying autocorrelation indicates inefficient exploration and reduces the effective sample size.

6.3 Gelman-Rubin Diagnostic ($\hat{R}$)

$\hat{R}$ evaluates stability by comparing **within-chain variability** to **between-chain variability**. Values of $\hat{R} \approx 1$ are desirable, suggesting chains behave similarly. Values noticeably above 1 (e.g., > 1.1) indicate instability and that chains have not yet converged to the target posterior.

R value of 5.5 suggests poor mixing or lack of convergence across chains. As a preliminary diagnostic, it suggests that the chains are not producing consistent sampling behavior for θ , so the posterior simulation for this parameter is not yet trustworthy. We would still want to examine other tools such as trace plots, density plots, effective sample size, and autocorrelation.

7 Diagnostic Remediation

If diagnostics show poor mixing, drift, or high autocorrelation:

- **Increase Warmup:** If early iterations are influenced by starting values.
- **Run Longer:** Increase iterations to improve N_{eff} and $\hat{R}$ stability.
- **Re-parameterize:** Revisit priors or model specification if problems persist across all diagnostics.